



MACHINE LEARNING

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CHAPTER 1

Introduction

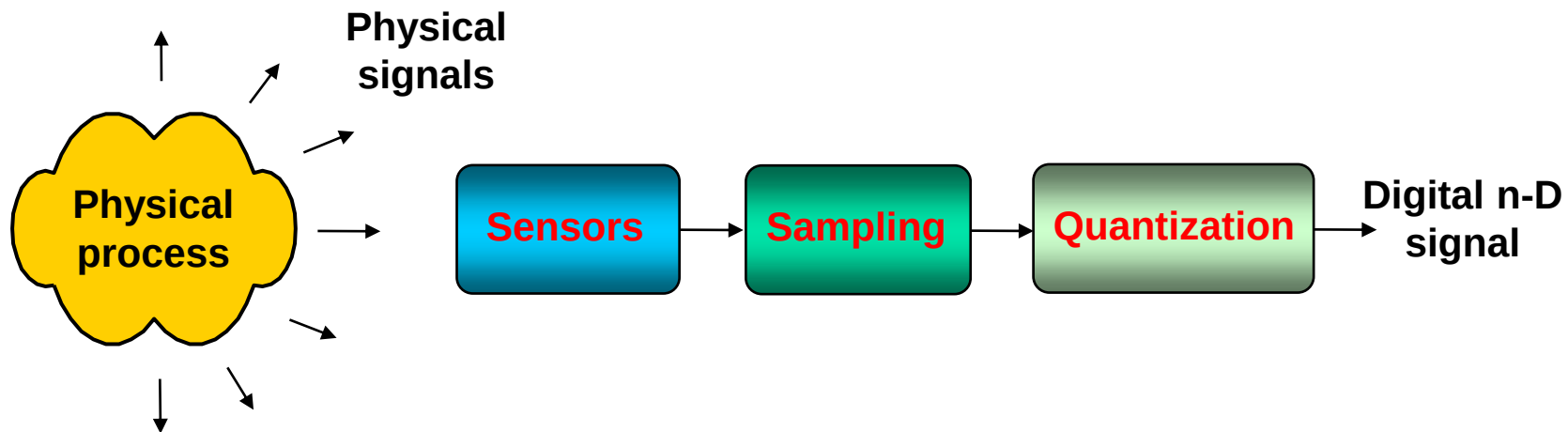
- Basic Concepts
- Machine Learning System
- Design Example
- Application Examples

Mono/Multidimensional Signals

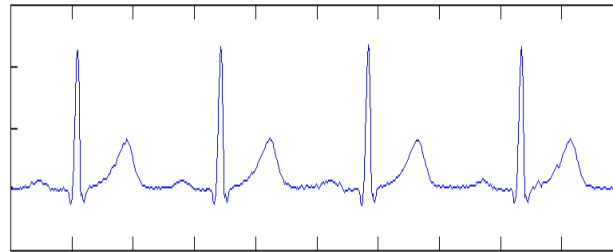
- **Signal**: a varying quantity that carries information about a physical phenomenon/process under analysis.
- **Monodimensional signal**: function $f(\xi)$ representing the information evolution with respect to a variable ξ which translates a **physical reality** such as time, frequency, pressure, etc...
- **Multidimensional signal**: information evolution is simultaneously related to multiple correlated or uncorrelated physical realities $\rightarrow g(\xi, \gamma, \delta, \dots)$

Nota: Physical realities may be continuous as well as discrete variables.

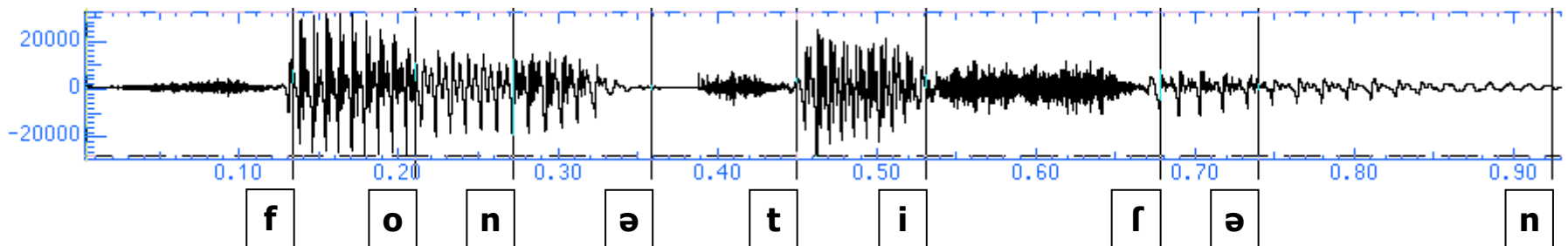
Signal Acquisition



Examples of 1-D Signals



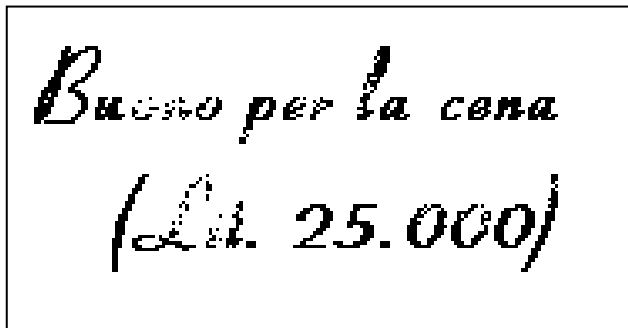
ECG signal



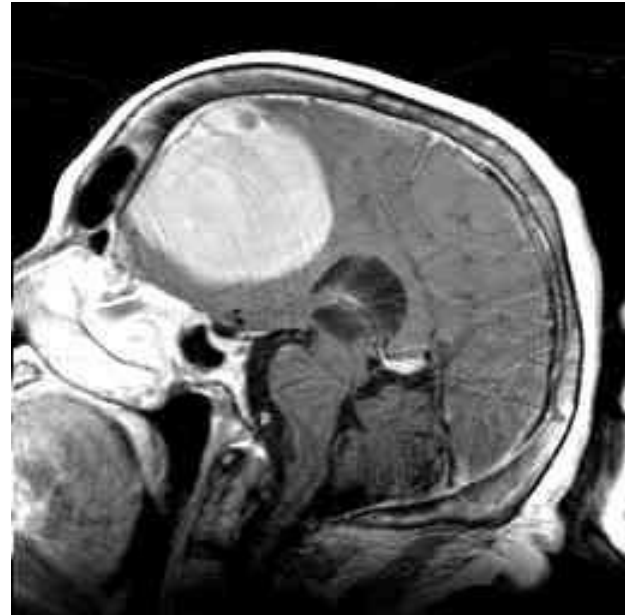
Speech Signal (oscillogram)

Phonetician

Examples of 2-D Signals

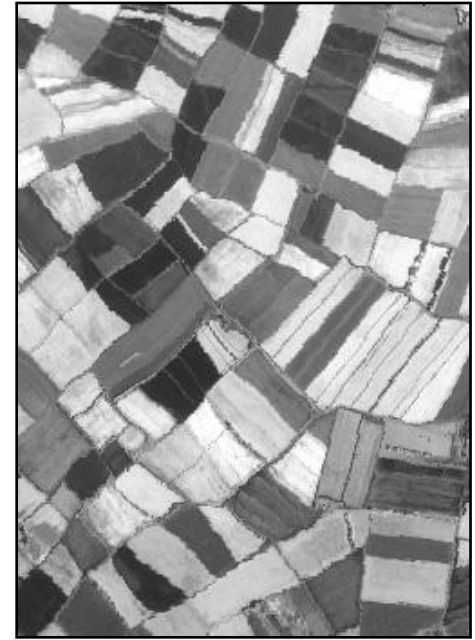
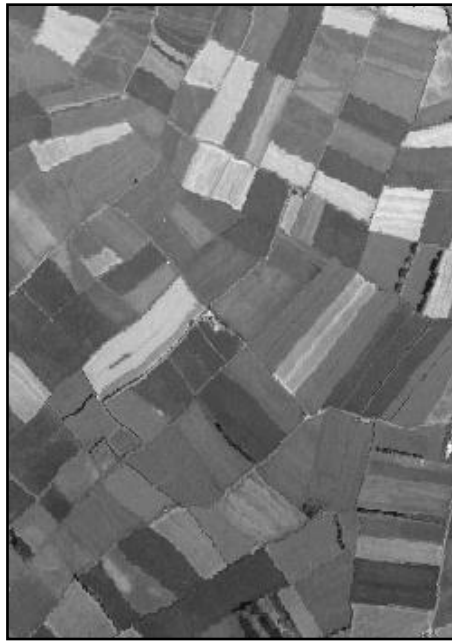
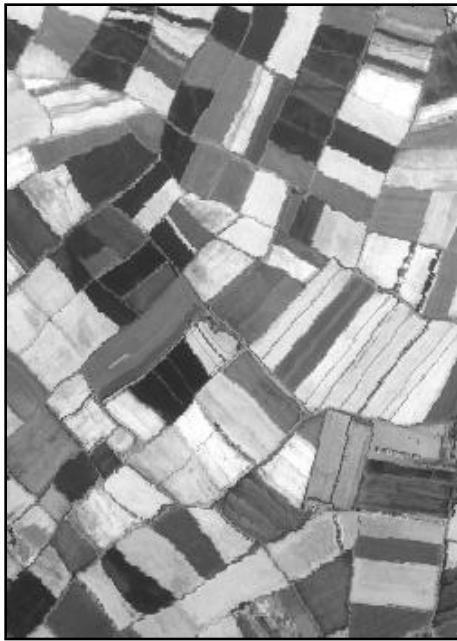


Black & White Image



Biomedical Image (10 bit)

Examples of n-D Signals



**Multispectral Remote Sensing
Image (8 bit, 3 bands)**

Learning

- **“Machine learning** is the study of computer algorithms that improve automatically through experience. It is seen as a subset of artificial intelligence. Machine learning algorithms build a mathematical model based on sample data, known as “training data”, in order to make predictions or decisions without being explicitly programmed to do so.” (from Wikipedia).
- **Supervised learning**: the learner is fed with a set of input/output pairs (**training set**).
- **Unsupervised learning**: no output information is provided, but just input data which are modeled by the learner (e.g., **data clustering**).

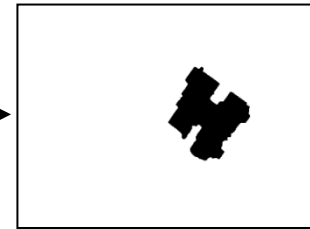
Learning

- **Semi-supervised learning**: besides a limited amount of input/output pairs, the plenty **unlabeled data** are exploited during the learning process.
- **Active learning**: the learner **interactively queries** a supervisor (**oracle**) to label new samples with the desired outputs.
- **Reinforcement learning**: learning process of an arbitrary being (**agent**) in the world surrounding it (**environment**). The agent seeks to **maximize the rewards** it receives from the environment, and performs different actions in order to learn how the environment responds and gain more rewards.

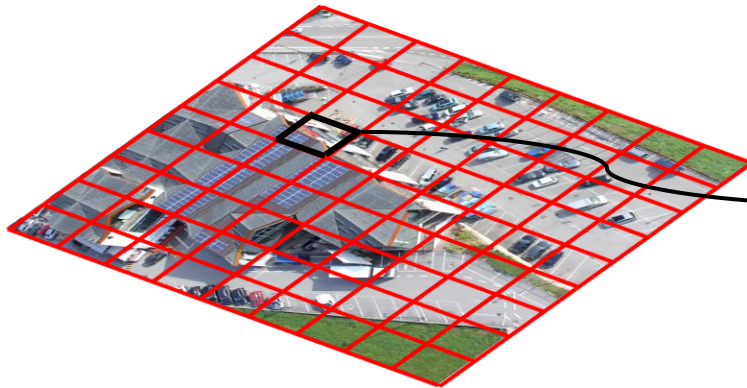
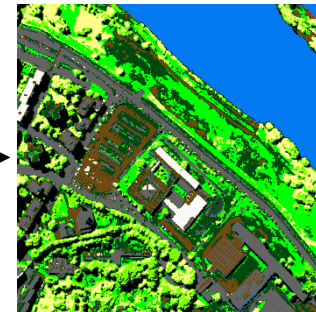
Supervised Tasks



**Binary
Classification**



**Multiclass
Classification**



**Multilabel
Classification**

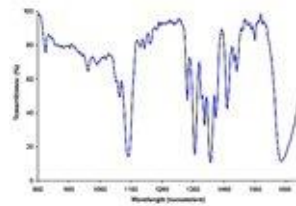
0	Asphalt
1	Solar panel
...	...
0	Tree
1	Car

Supervised Tasks



Captioning

Eight cars on the right in a parking lot.



Regression

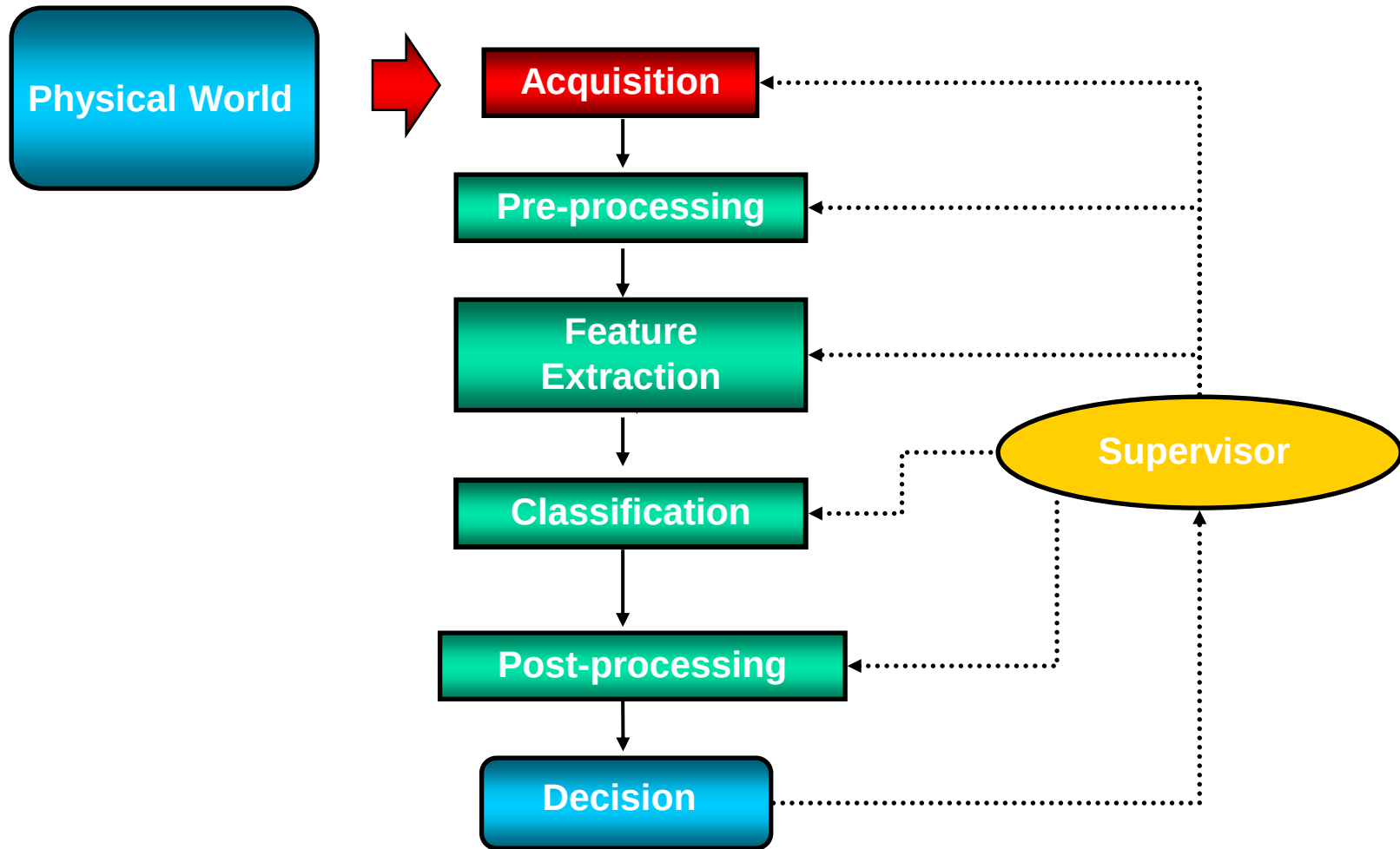
Saccharose: 23.65 [g/l]

First Last	
The following table contains a resume for a person who has worked in the medical field. Apply your knowledge, training and skills to the person's background and provide a ranking for the person's background. (10 points)	
EDUCATION	PROFESSIONAL EXPERIENCE
- State University - State of Illinois - State of Illinois - Local Community College - Community College - State of Illinois - State of Illinois	- DATA SUPPORT ANALYST at large health care company - City, State and Country 1. The person has worked in the medical field for 10 years. 2. The person has worked in the medical field for 10 years. 3. The person has worked in the medical field for 10 years. 4. The person has worked in the medical field for 10 years. 5. The person has worked in the medical field for 10 years. 6. The person has worked in the medical field for 10 years. 7. The person has worked in the medical field for 10 years. 8. The person has worked in the medical field for 10 years. 9. The person has worked in the medical field for 10 years. 10. The person has worked in the medical field for 10 years.
SKILLS	PATIENT ACCOUNT SPECIALIST at large health care company
- Business Analytics Support - Healthcare Analytics Support - Healthcare Analytics Support - Healthcare Analytics Support - Healthcare Analytics Support - Healthcare Analytics Support - Healthcare Analytics Support - Healthcare Analytics Support - Healthcare Analytics Support - Healthcare Analytics Support	1. The person has worked in the medical field for 10 years. 2. The person has worked in the medical field for 10 years. 3. The person has worked in the medical field for 10 years. 4. The person has worked in the medical field for 10 years. 5. The person has worked in the medical field for 10 years. 6. The person has worked in the medical field for 10 years. 7. The person has worked in the medical field for 10 years. 8. The person has worked in the medical field for 10 years. 9. The person has worked in the medical field for 10 years. 10. The person has worked in the medical field for 10 years.
AFFILIATIONS/INTERESTS	BILLING DEPARTMENT LEAD at large health care company
- State of Illinois - State of Illinois - State of Illinois - State of Illinois - State of Illinois - State of Illinois - State of Illinois - State of Illinois - State of Illinois - State of Illinois	1. The person has worked in the medical field for 10 years. 2. The person has worked in the medical field for 10 years. 3. The person has worked in the medical field for 10 years. 4. The person has worked in the medical field for 10 years. 5. The person has worked in the medical field for 10 years. 6. The person has worked in the medical field for 10 years. 7. The person has worked in the medical field for 10 years. 8. The person has worked in the medical field for 10 years. 9. The person has worked in the medical field for 10 years. 10. The person has worked in the medical field for 10 years.

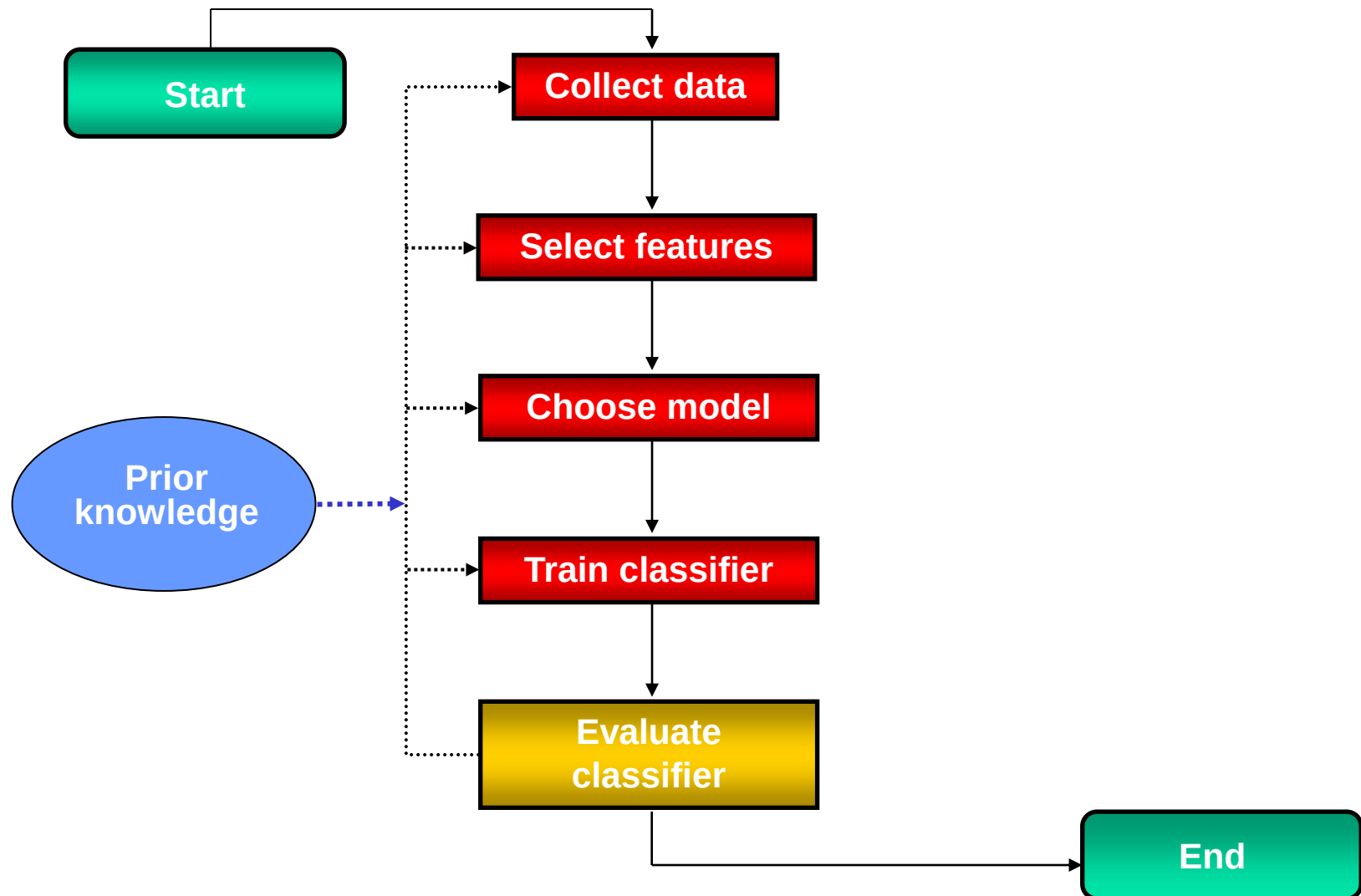
Ranking

Priority 1

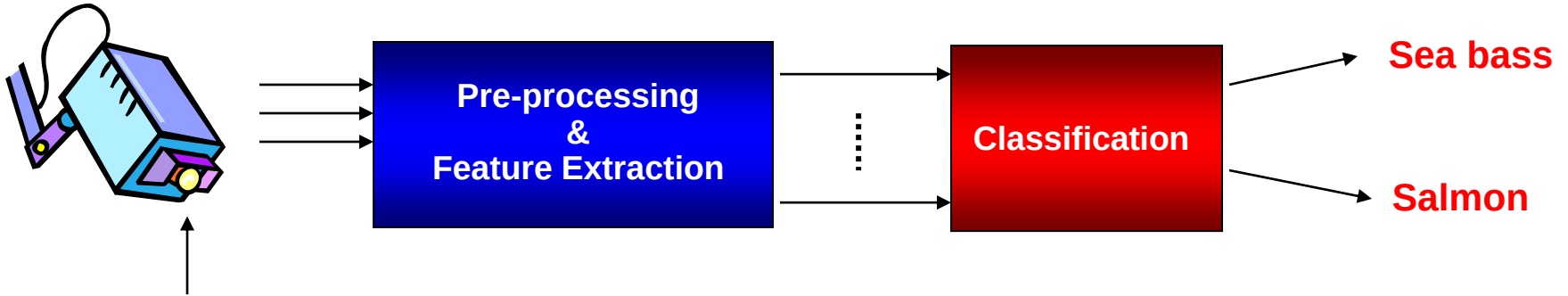
ML System: Block Scheme



ML System: Design Phases



Example: Automatic Fish-Packing Plant

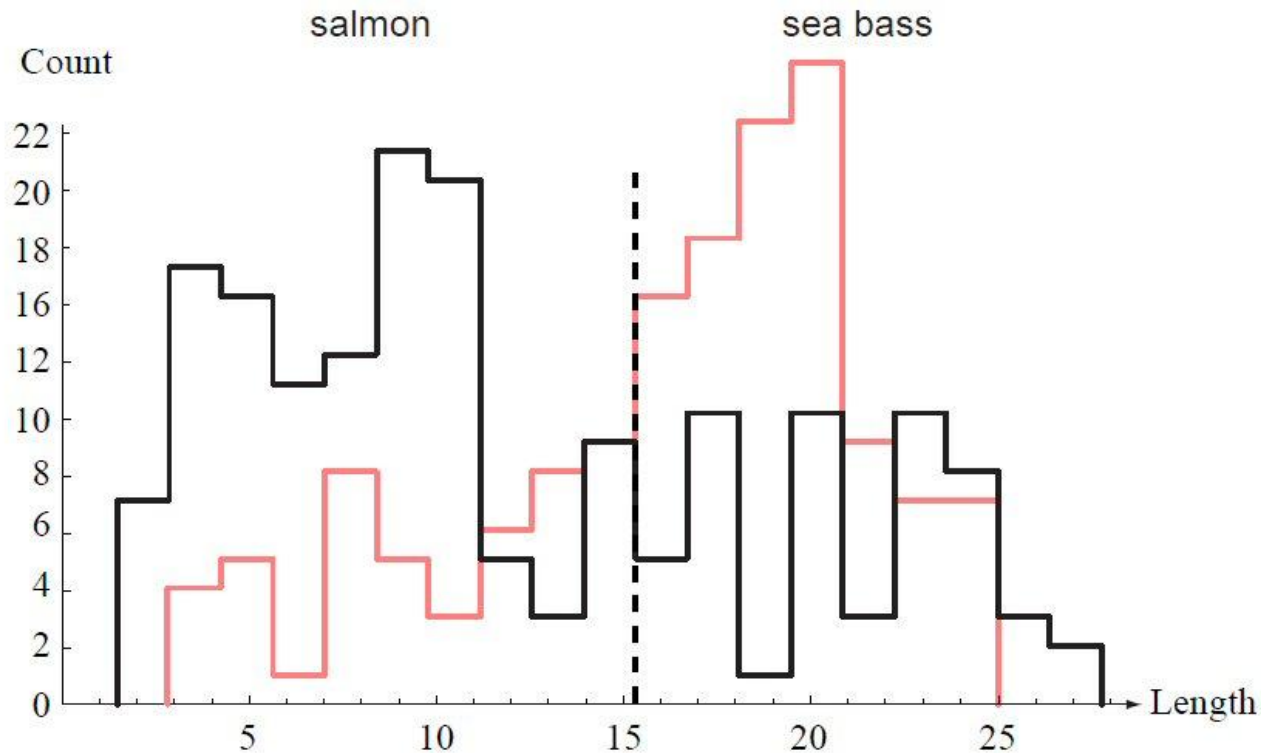


Example: Automatic Fish-Packing Plant

- **Pre-processing**: apply a segmentation operation in order to isolate fishes from one another and from the background.
- **Feature extraction**: measure some features or properties from the image which will help in discriminating the two species of fish considered (e.g., fish length and width).
- **Classification**: evaluate the evidence presented and make a final decision as to the species.

Example: Automatic Fish-Packing Plant

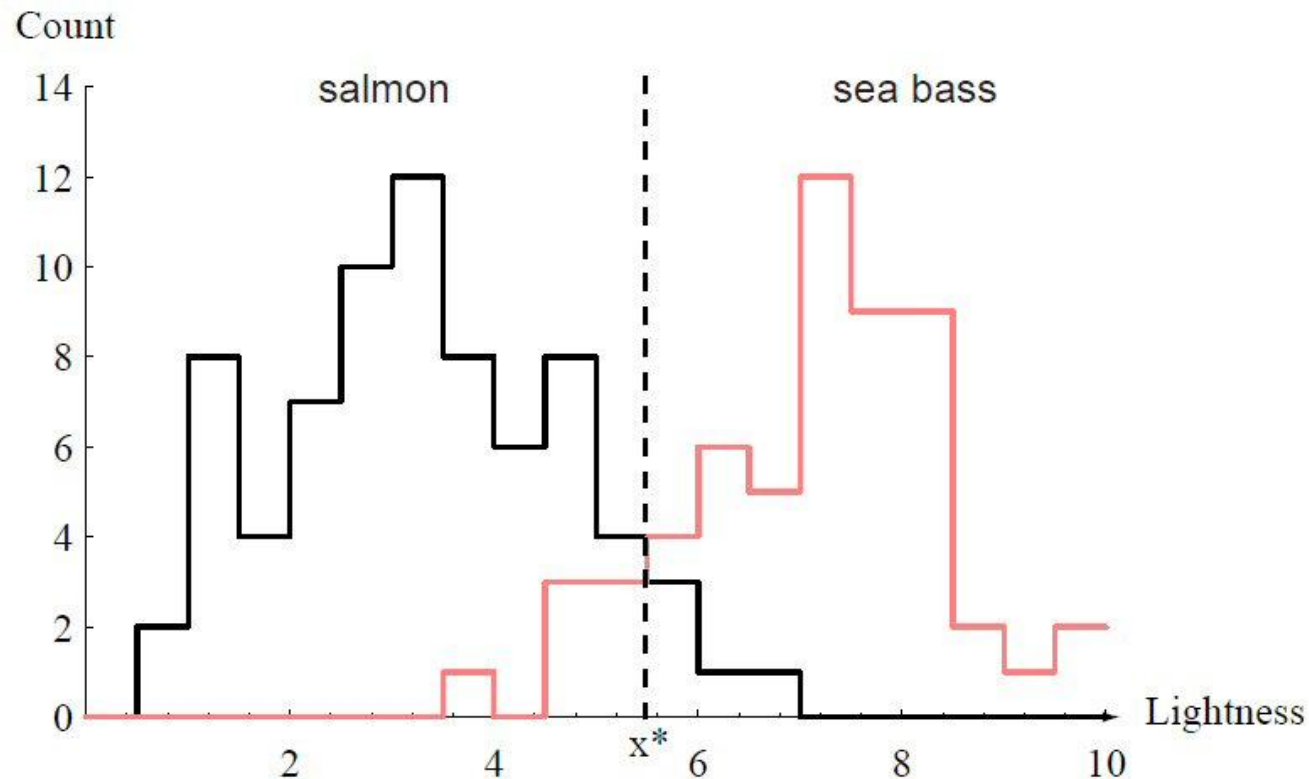
Classification Based on Length Feature



Using length appears not effective!

Example: Automatic Fish-Packing Plant

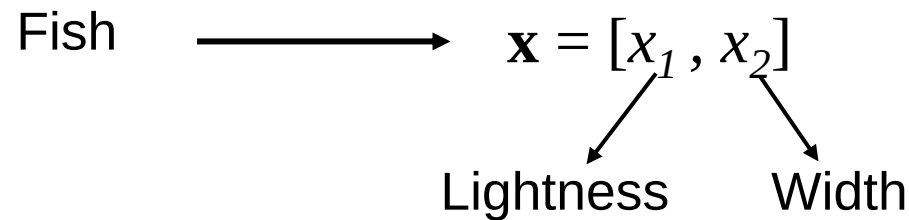
Classification Based on Lightness Feature



Looks better but still not enough satisfactory.

Example: Automatic Fish-Packing Plant

- In order to increase further the **discrimination capability**, one may think to adopt **two features** instead of just one.
- For instance, one could consider the width feature in addition to the lightness one:



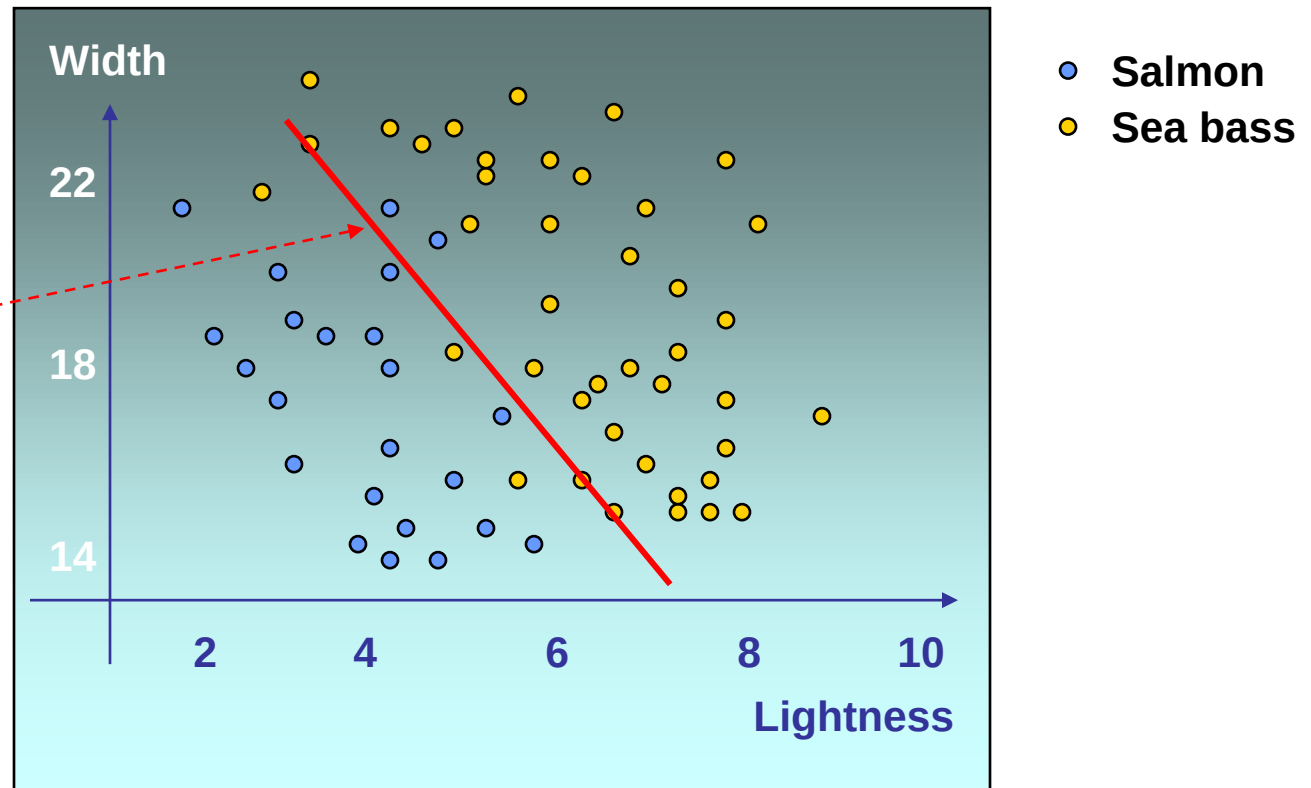
- **Problem** $\longrightarrow$ we need to define:
 - **classification model**
 - **cost function** (e.g., minimize the number of errors)
- Let us first consider a linear classifier...

Example: Automatic Fish-Packing Plant

2D Feature Space – Linear Classifier

During the training phase, the classifier should estimate the best linear separation according to the adopted cost function.

Can we do better?



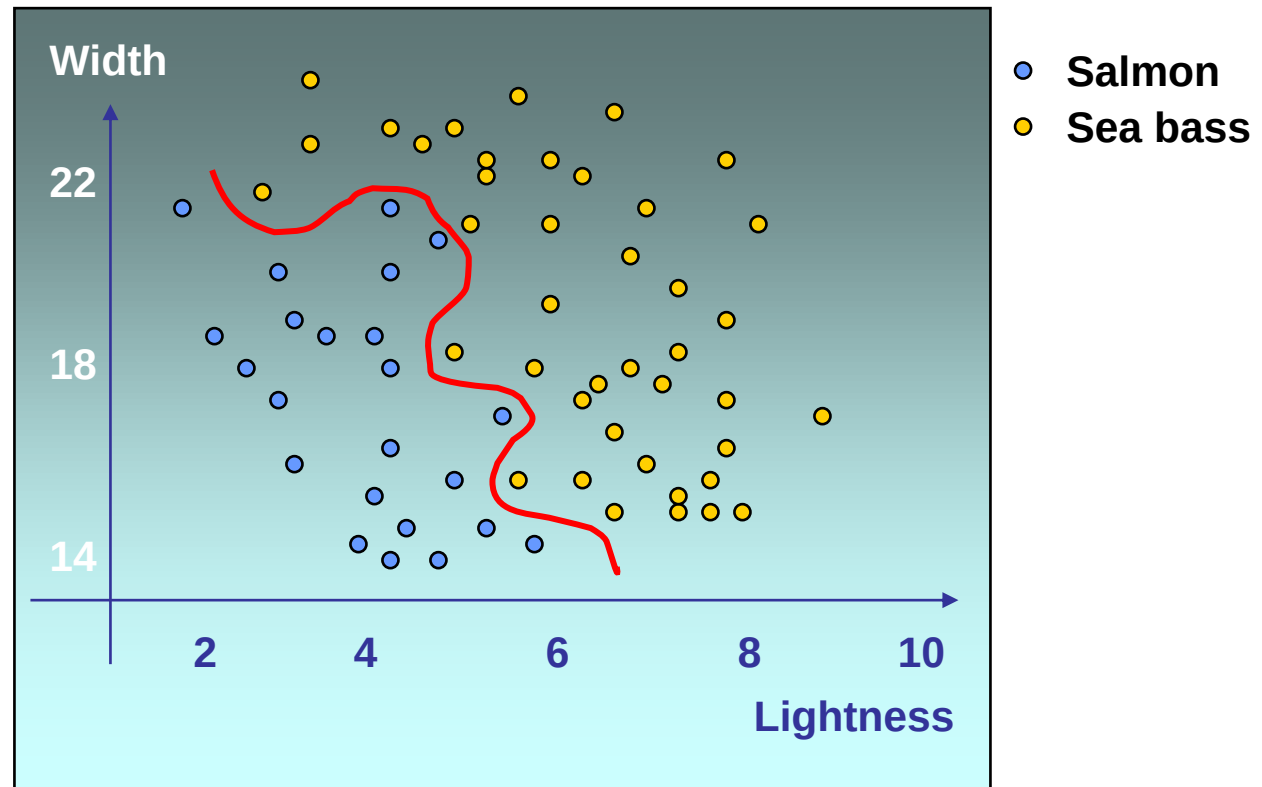
Example: Automatic Fish-Packing Plant

2D Feature Space – Highly Nonlinear Classifier

What about generalization?



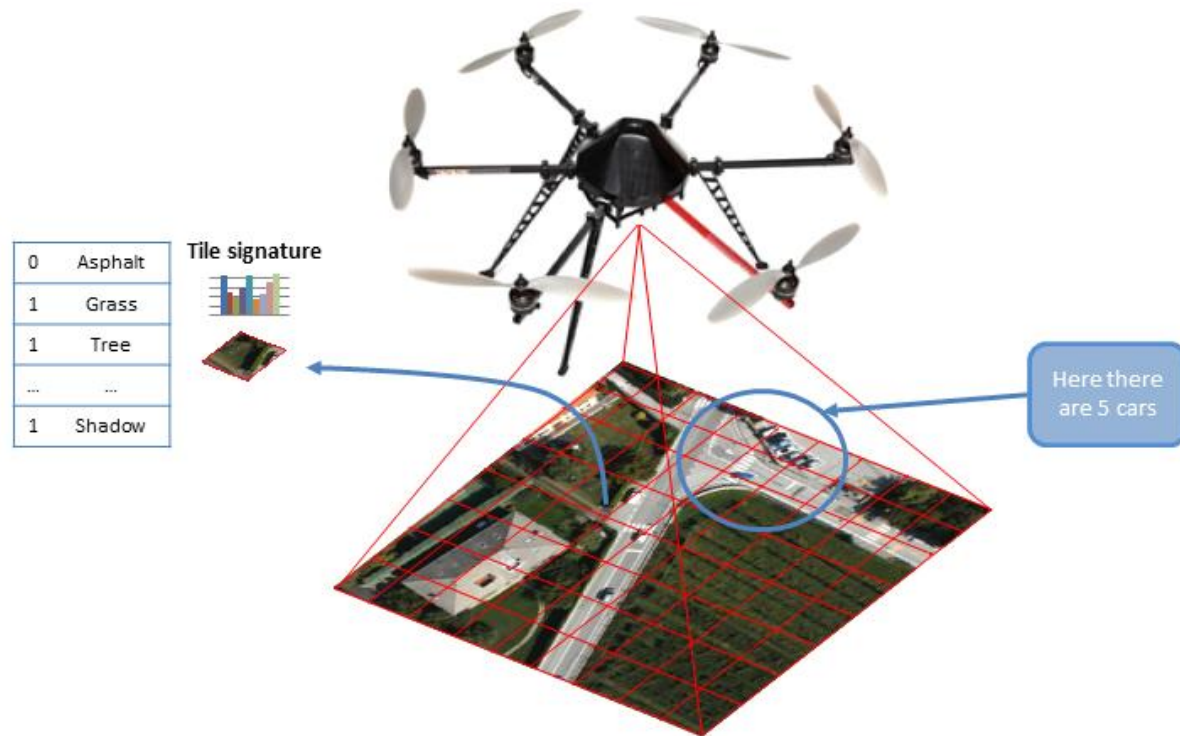
It is important to not tune excessively (**overfit**) the classifier on the training samples so that to leave room for a correct (at least satisfactory) classification of **novel patterns** (samples)!



Application Domains



Example: UAV Image Analysis



Example: UAV Image Analysis



Grass



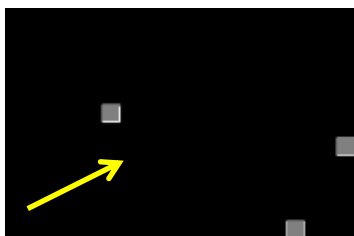
Tree



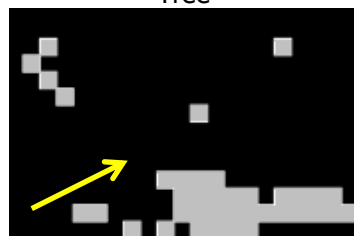
Asphalt



Cars



Solar Panel



Vineyard



Roof1



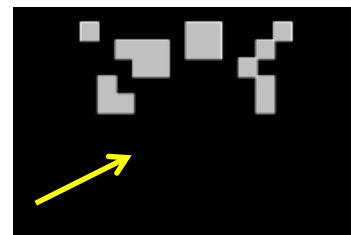
Roof2



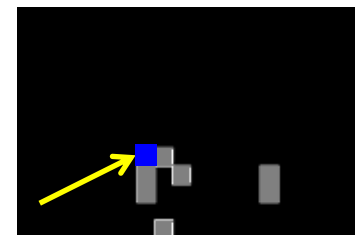
Soil



People



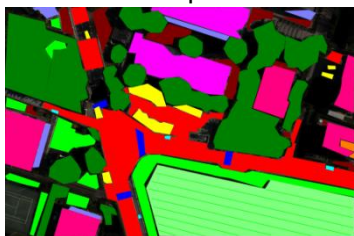
Building Facade



Pedestrian Crossing



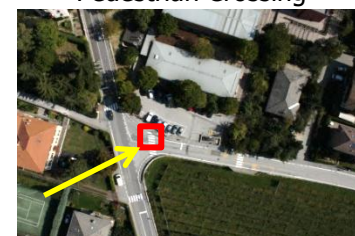
Shadow



Ground-Truth

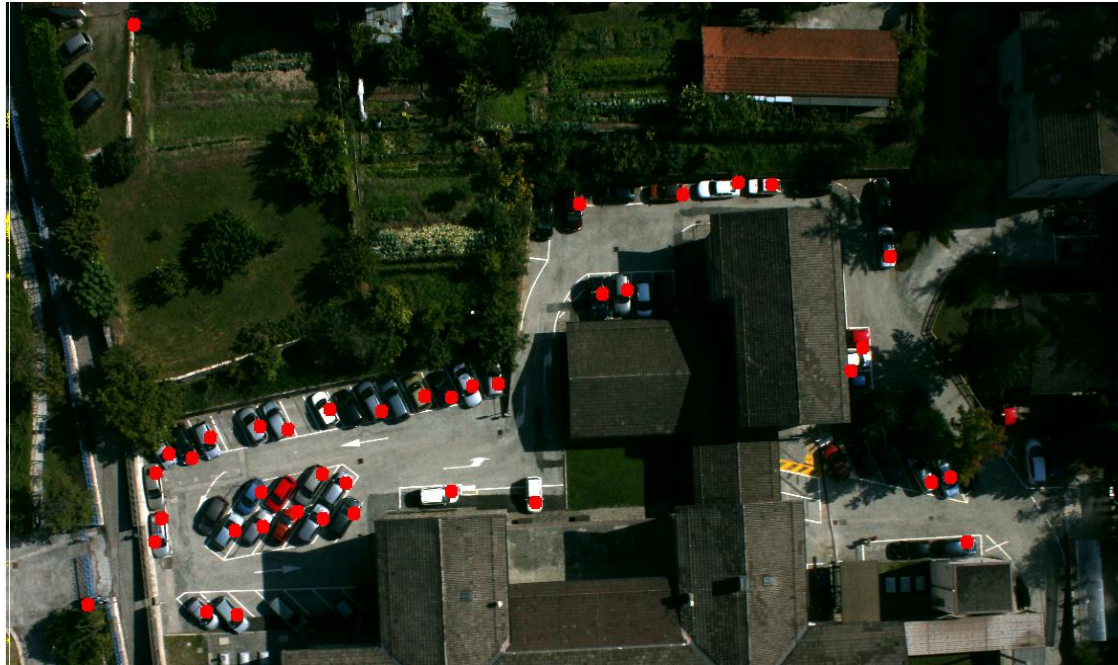


Pixel-based classification

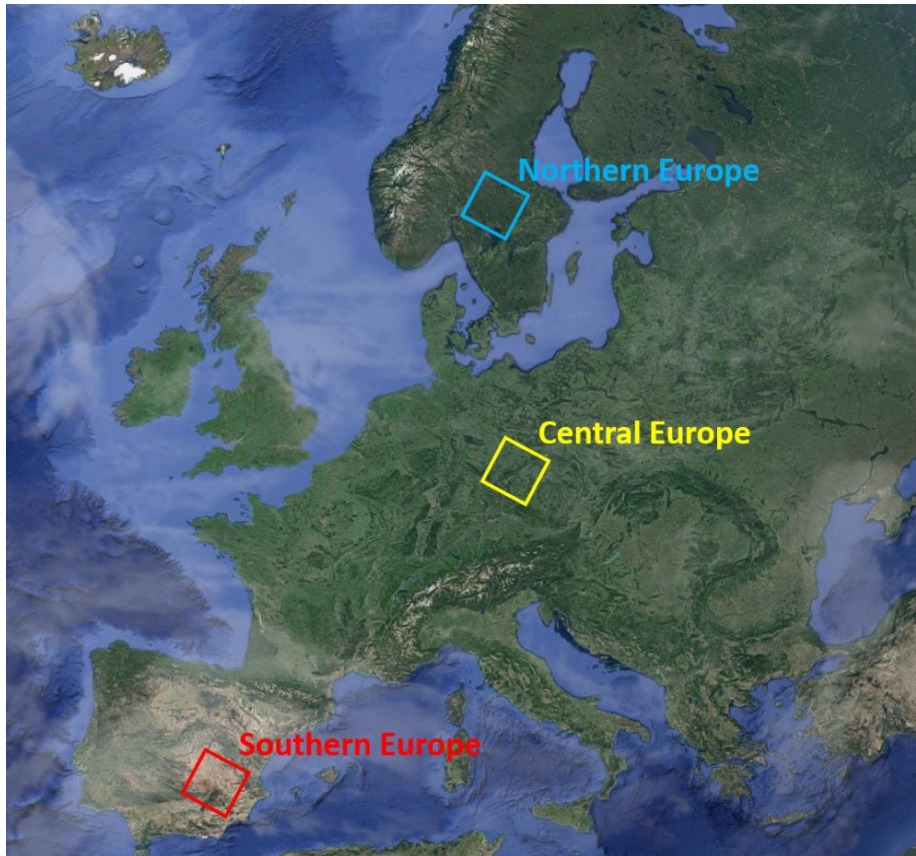


Original Image

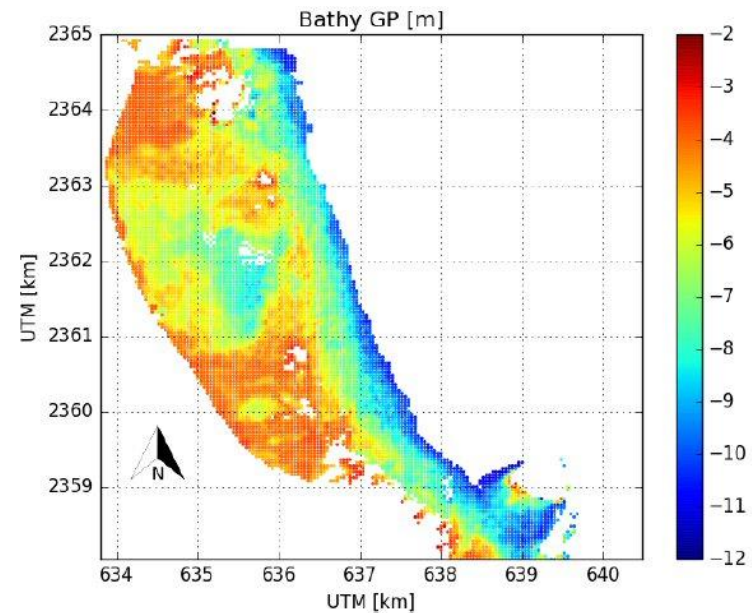
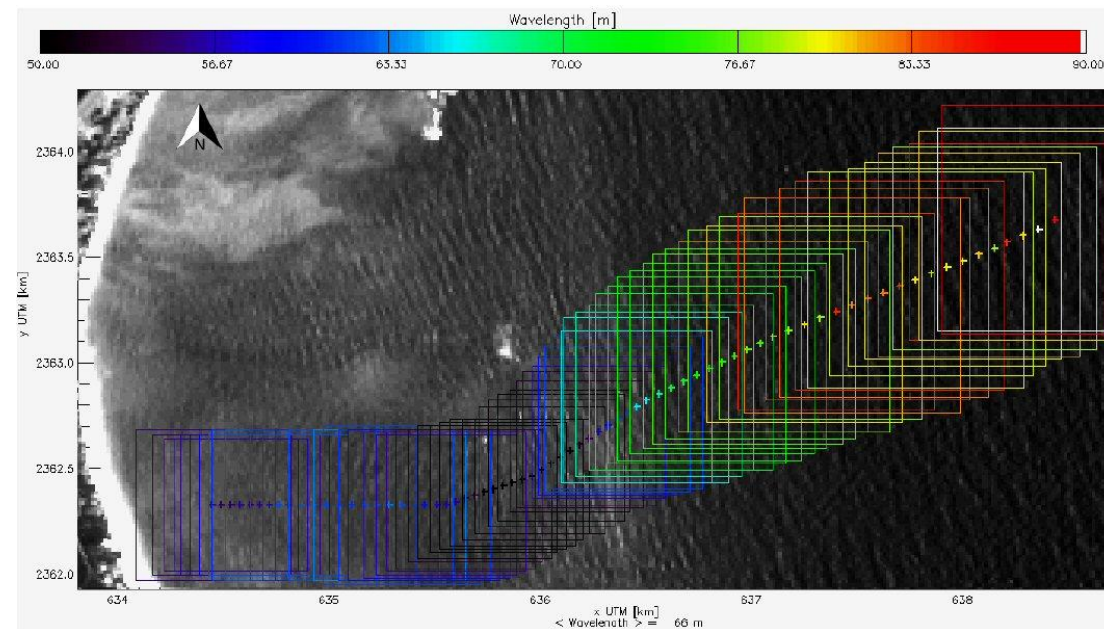
Example: UAV Image Analysis



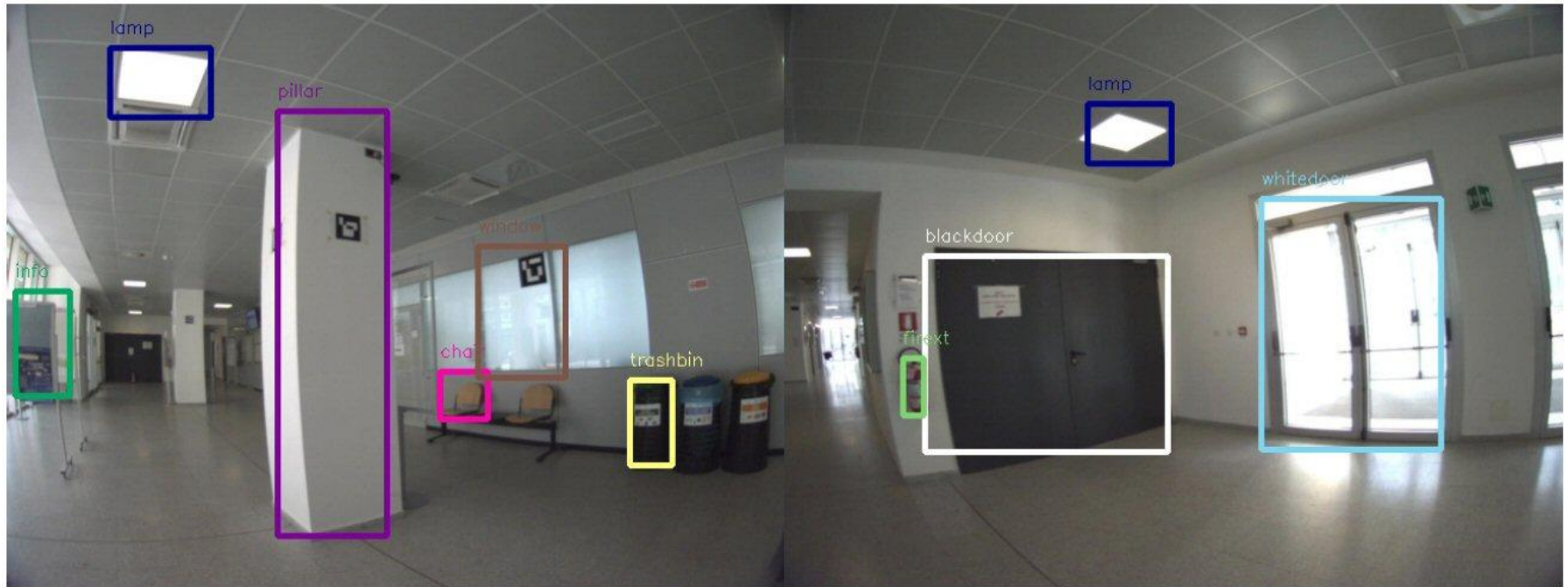
Example: Large Scale Mapping



Example: Bathymetry Mapping



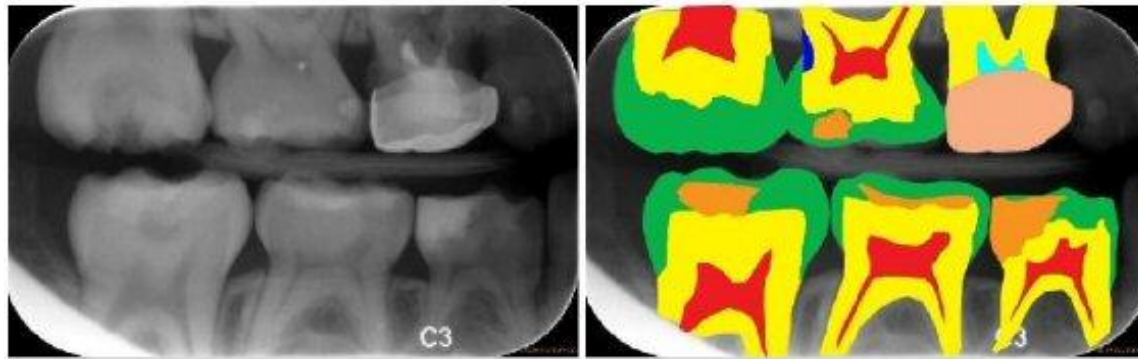
Example: Object Detection



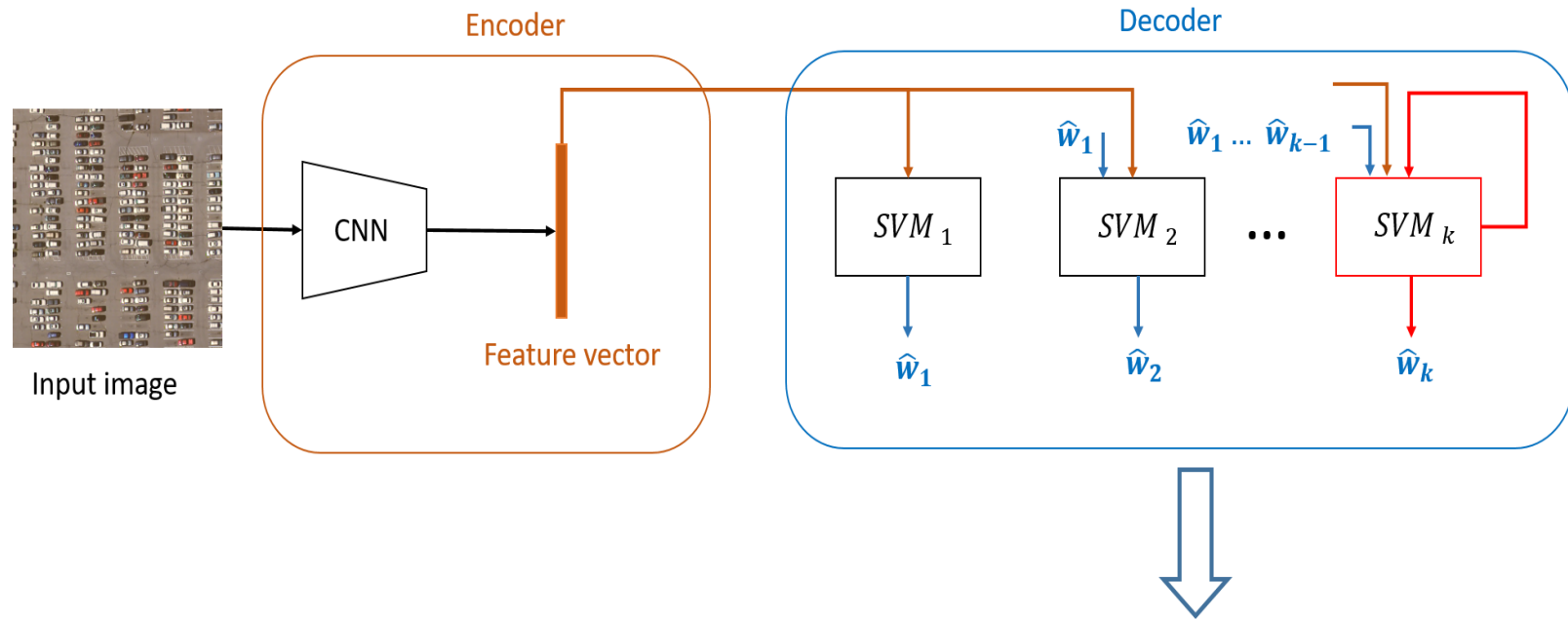
Example: Event Recognition



Example: Semantic Segmentation

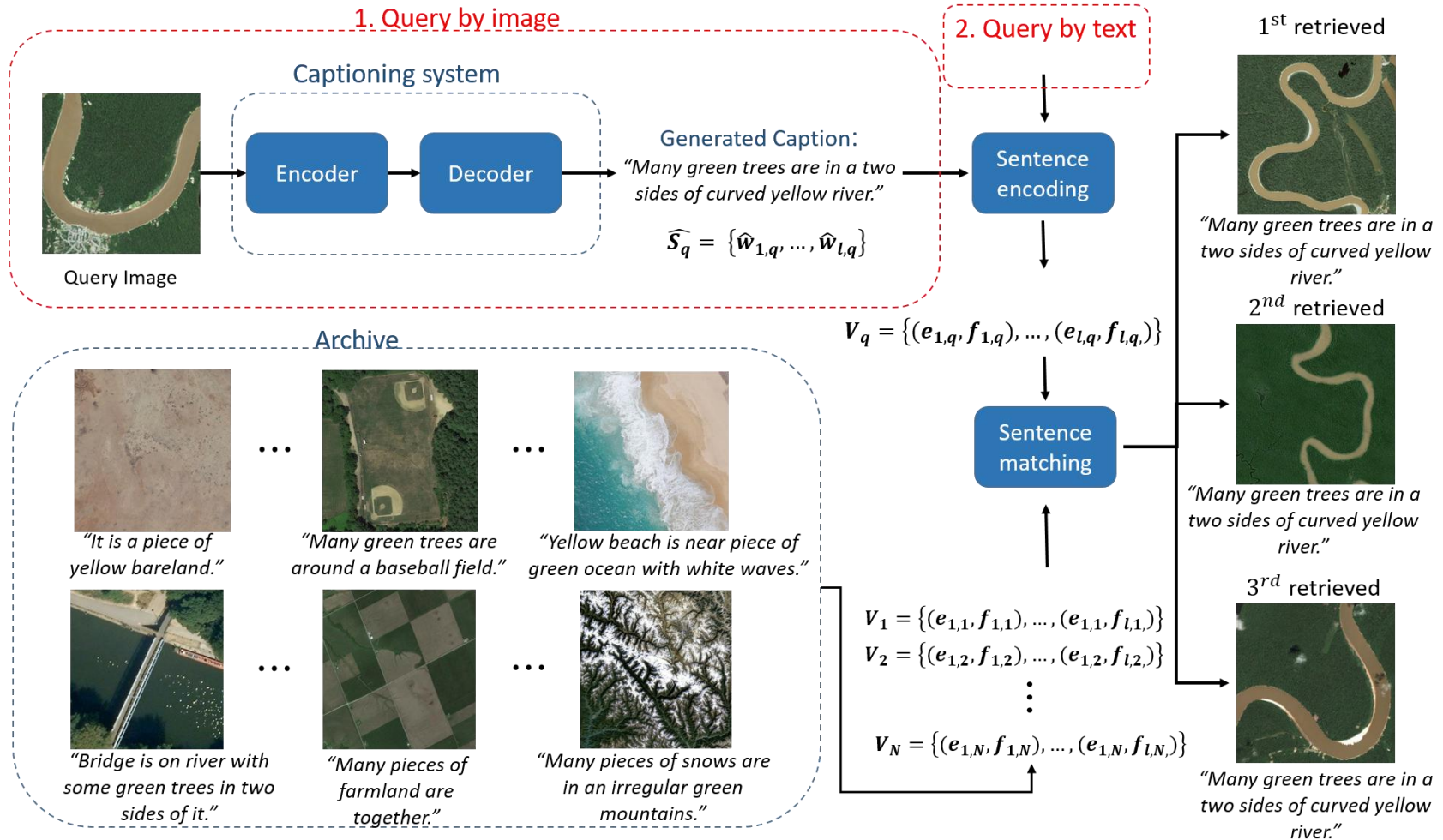


Example: Image Captioning

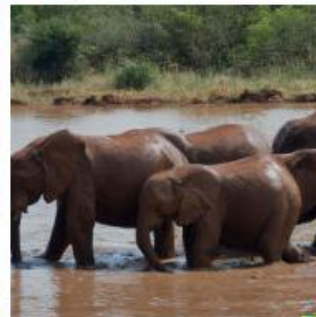
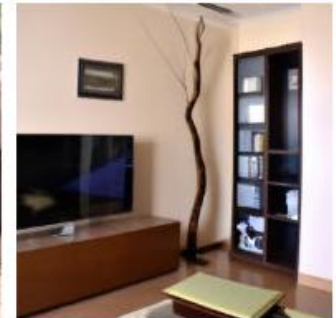


Generated description: Lots of cars parked closely with many empty parking spaces in the parking lot.

Example: Image Retrieval



Example: Text-to-Image Synthesis



“a green train is coming down the tracks”

“a group of skiers are preparing to ski down a mountain.”

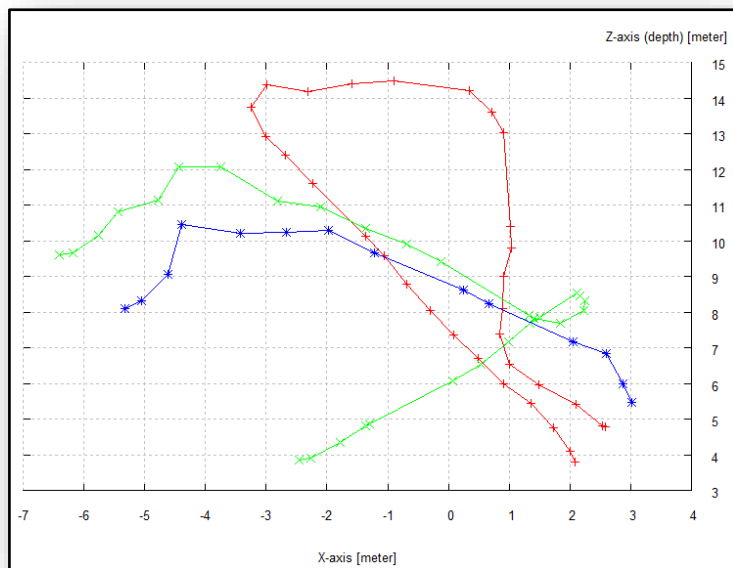
“a small kitchen with a low ceiling”

“a group of elephants walking in muddy water.”

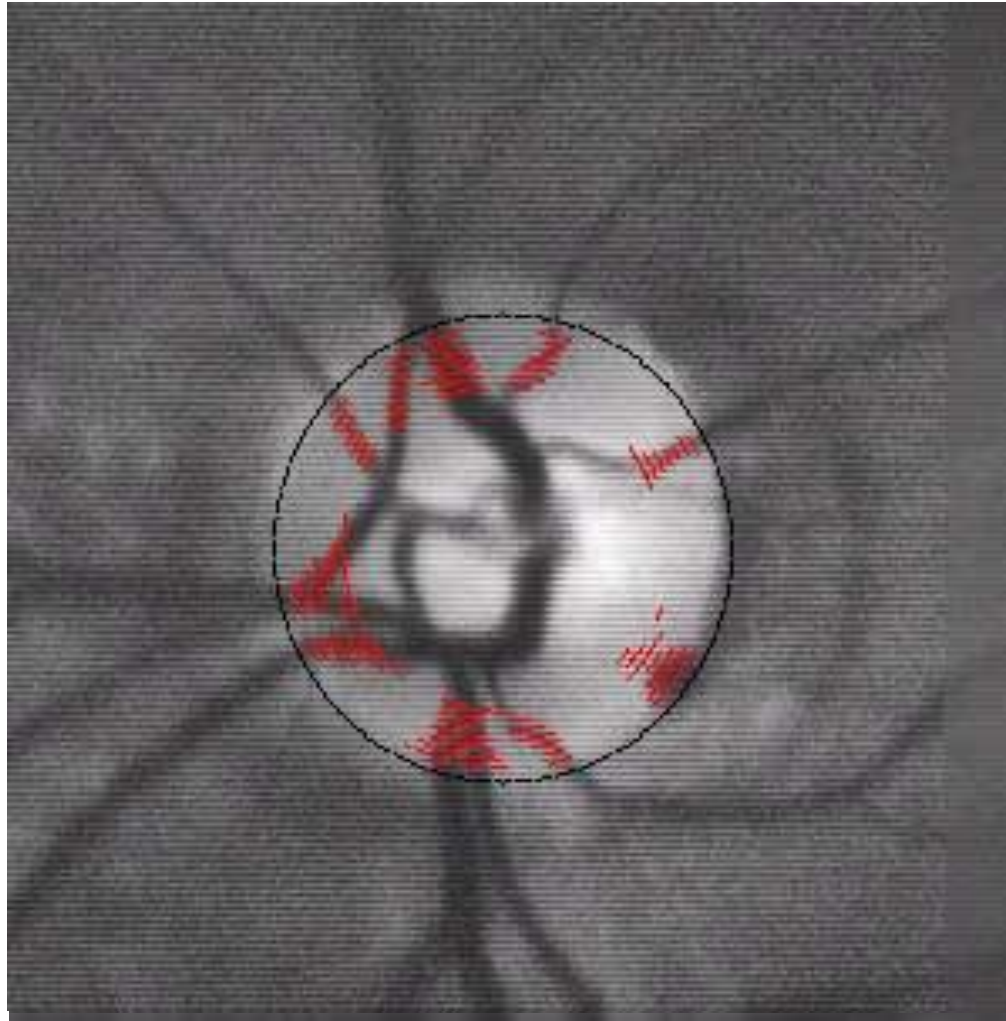
“a living area with a television and a table”

Example: Video Surveillance

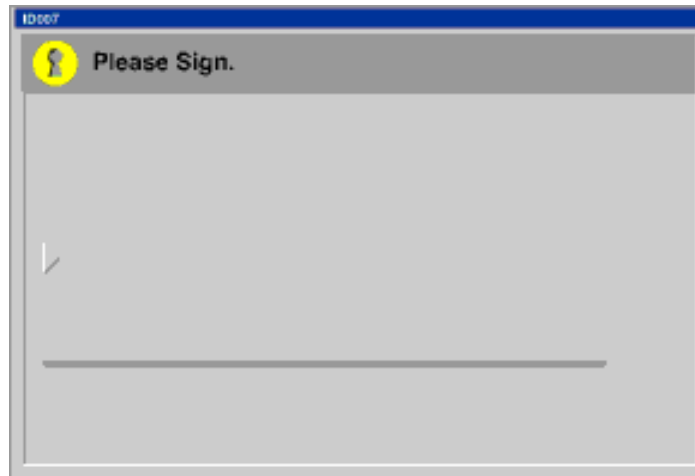
- ❑ Multiple person tracking
- ❑ Behavioral Analysis



Example: Biometry

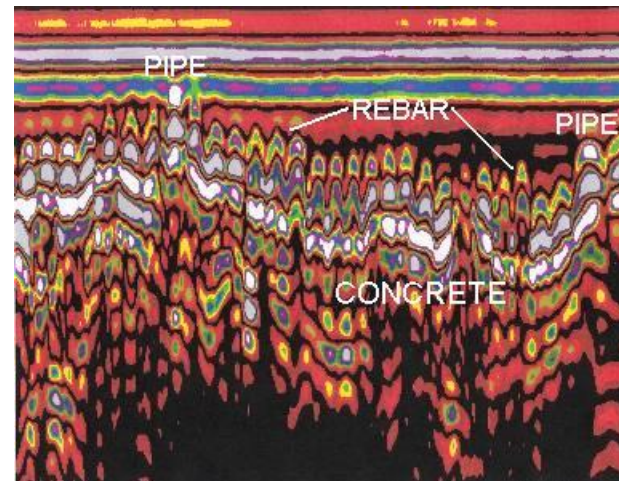


Example: Biometry

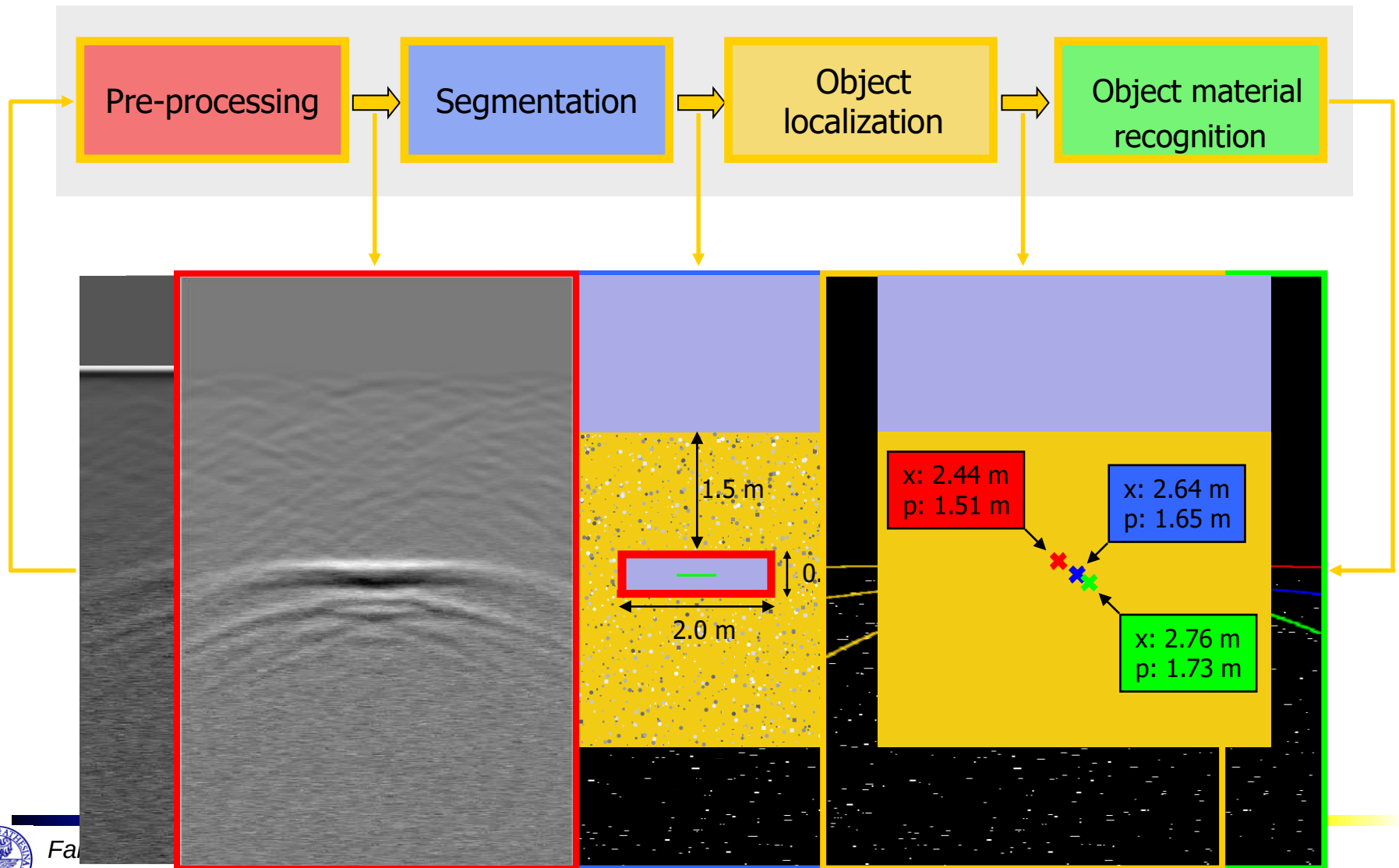


Example: Ground Penetrating Radar

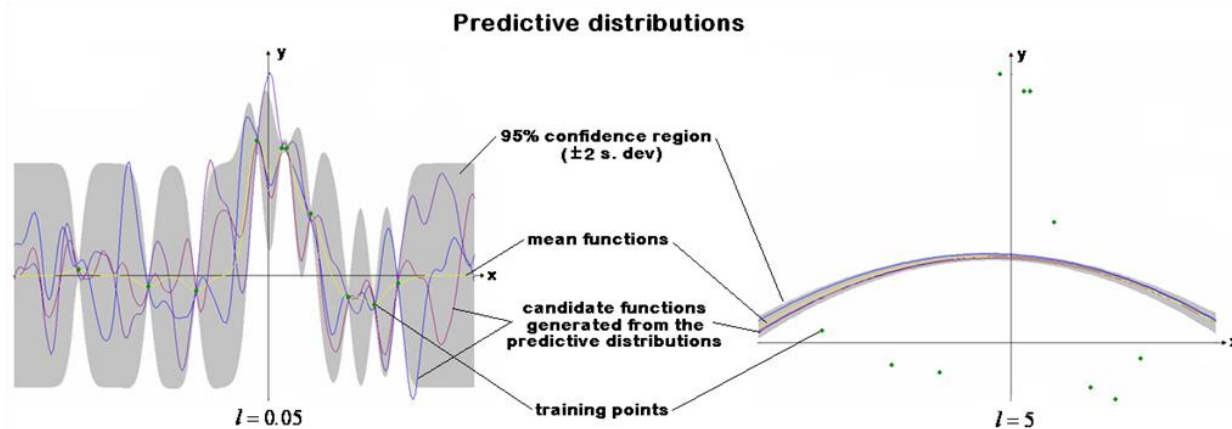
- Underground Storage Tanks
- Concrete Inspection
- Landfills and Burial Trenches
- Archaeological Studies
- Underground Structures



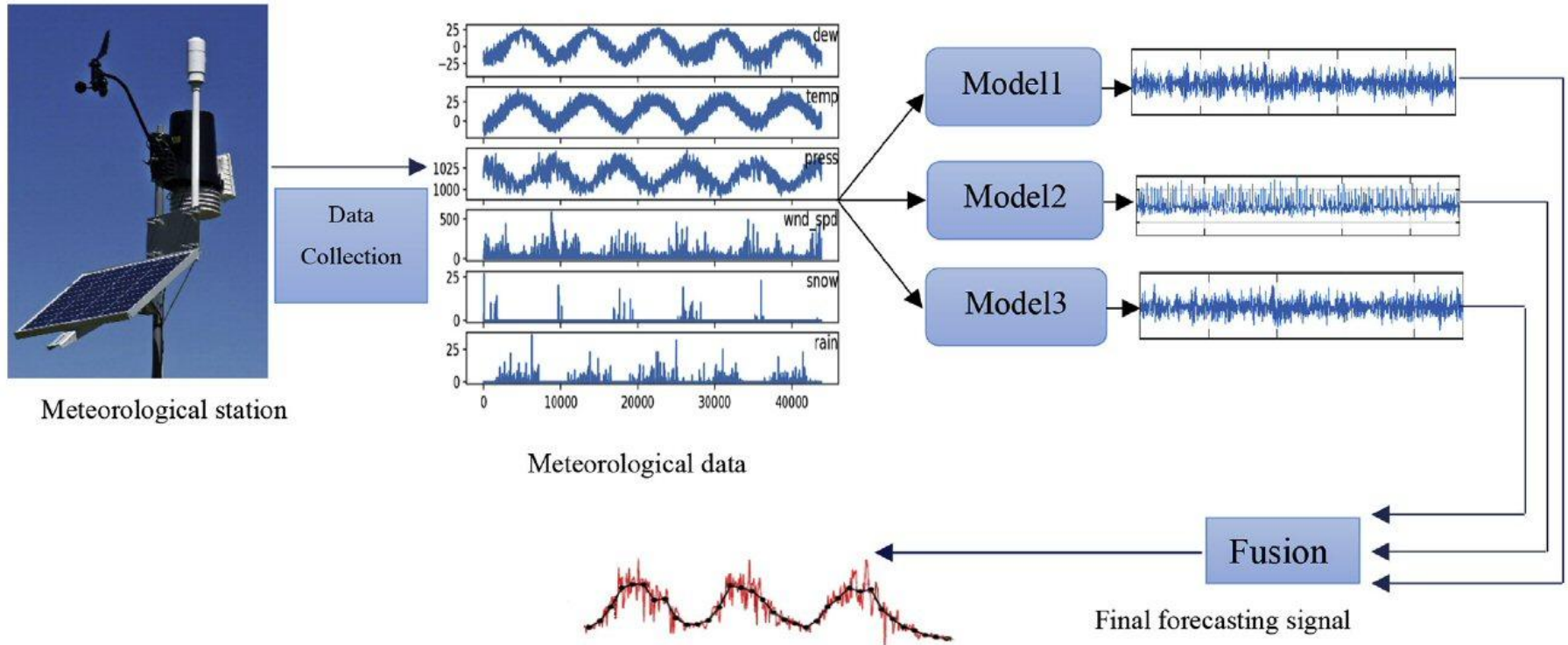
Example: Ground Penetrating Radar



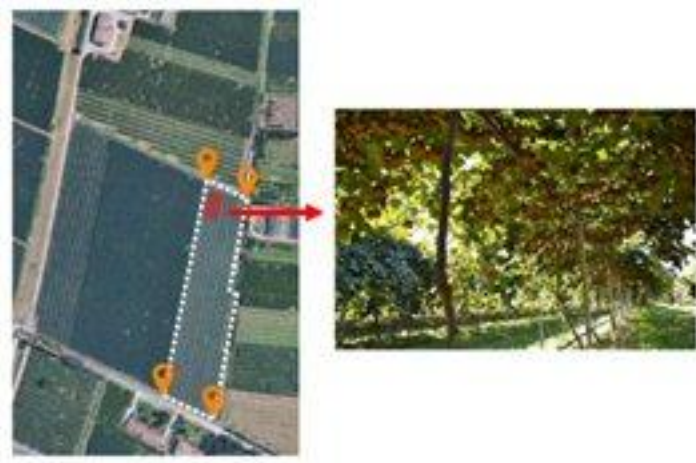
Example: Chemometric Data Analysis



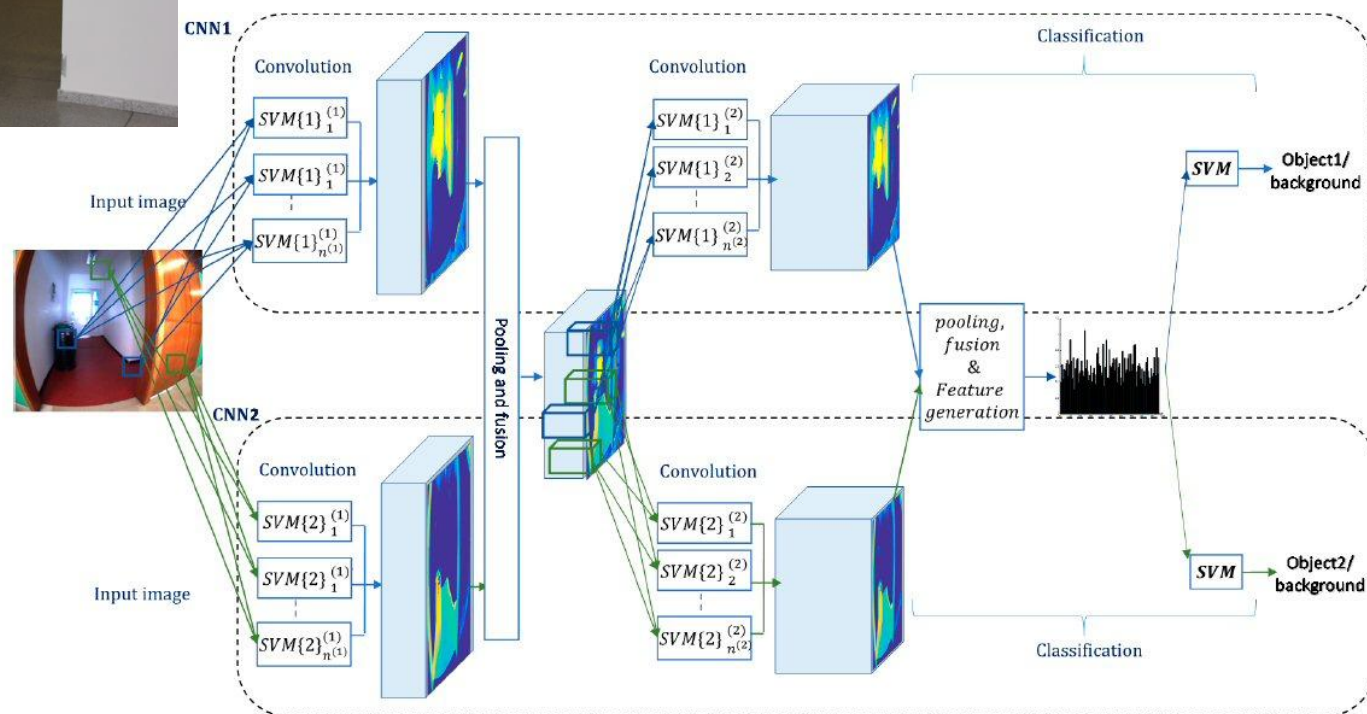
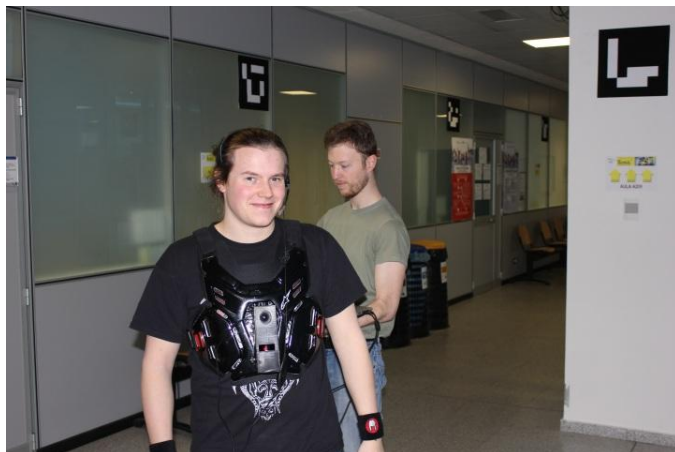
Example: Solar Radiation Forecasting



Example: Agriculture



Example: Assistive Technologies



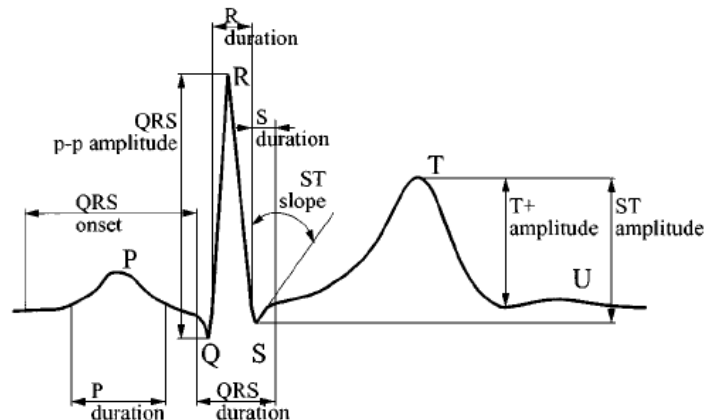
Example: Cardiac Pathology Detection



Heart

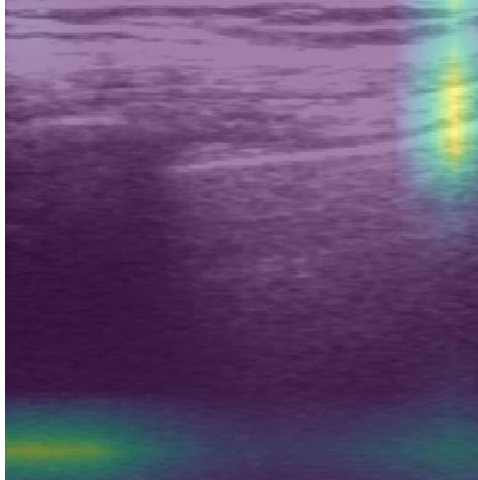


ECG signal

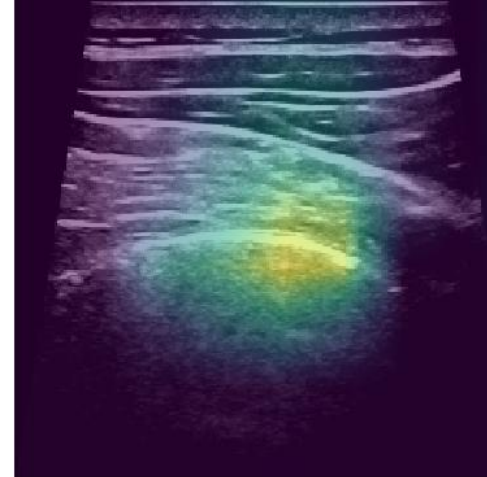


Example: COVID-19 Detection

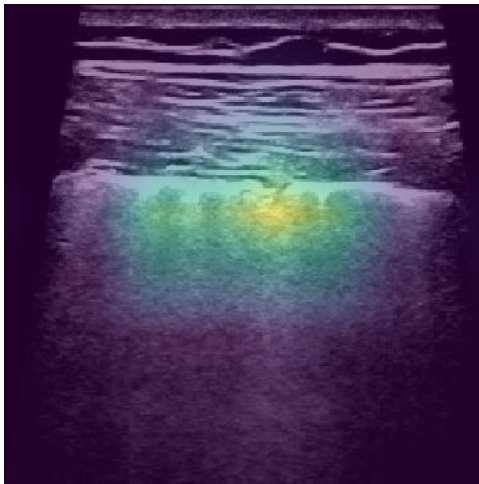
Case 0



Case 1



Case 2



Case 3

